# 1.2.25

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# Question

Let $x(t)$ be a continuous-time, real-valued signal band-limited to $F$ Hz. The Nyquist sampling rate, in Hz, for $y(t) = x(0.5t) + x(t) - x(2t)$ is

1. $F$
2. $2F$
3. $4F$
4. $8F$

# Definition of Nyquist Rate

Given Bandwidth of the signal is $F$.
Nyquist rate $= 2 \times$ highest frequency component of the signal.
Fourier transform of a scaled signal will be:

$$x(at) \leftarrow \mathcal{F} \rightarrow \frac{1}{|a|} X\left(\frac{f}{a}\right) \tag{1}$$

$$X(f) = 0 \quad \text{for} |f| > F \tag{2}$$

# Bandwidth of scaled signals

For $x\left(0.5t\right)$
Here a= 0.5, from eq. (1)

$$x(0.5t) \leftarrow \mathcal{F} \rightarrow \frac{1}{|0.5|} \tag{3}$$

$$\text{from eq (2)} \tag{4}$$

$$|2f| \leq F \Rightarrow |f| \leq \frac{F}{2} \tag{5}$$

Bandwidth $= 0.5F$

# Bandwidth of scaled signals

For $x\left(2t\right)$
Here a= 2, from eq. (1)

$$x(2t) \leftarrow \mathcal{F} \rightarrow \frac{1}{|2|} \tag{6}$$

$$\text{from eq (2)} \tag{7}$$

$$|0.5f| \leq F \Rightarrow |f| \leq 2F \tag{8}$$

Bandwidth $= 2F$ For $x(t)$ bandwidth will be same.

# Bandwidth of combined signal

It will be the maximum of the of all bandwidths.

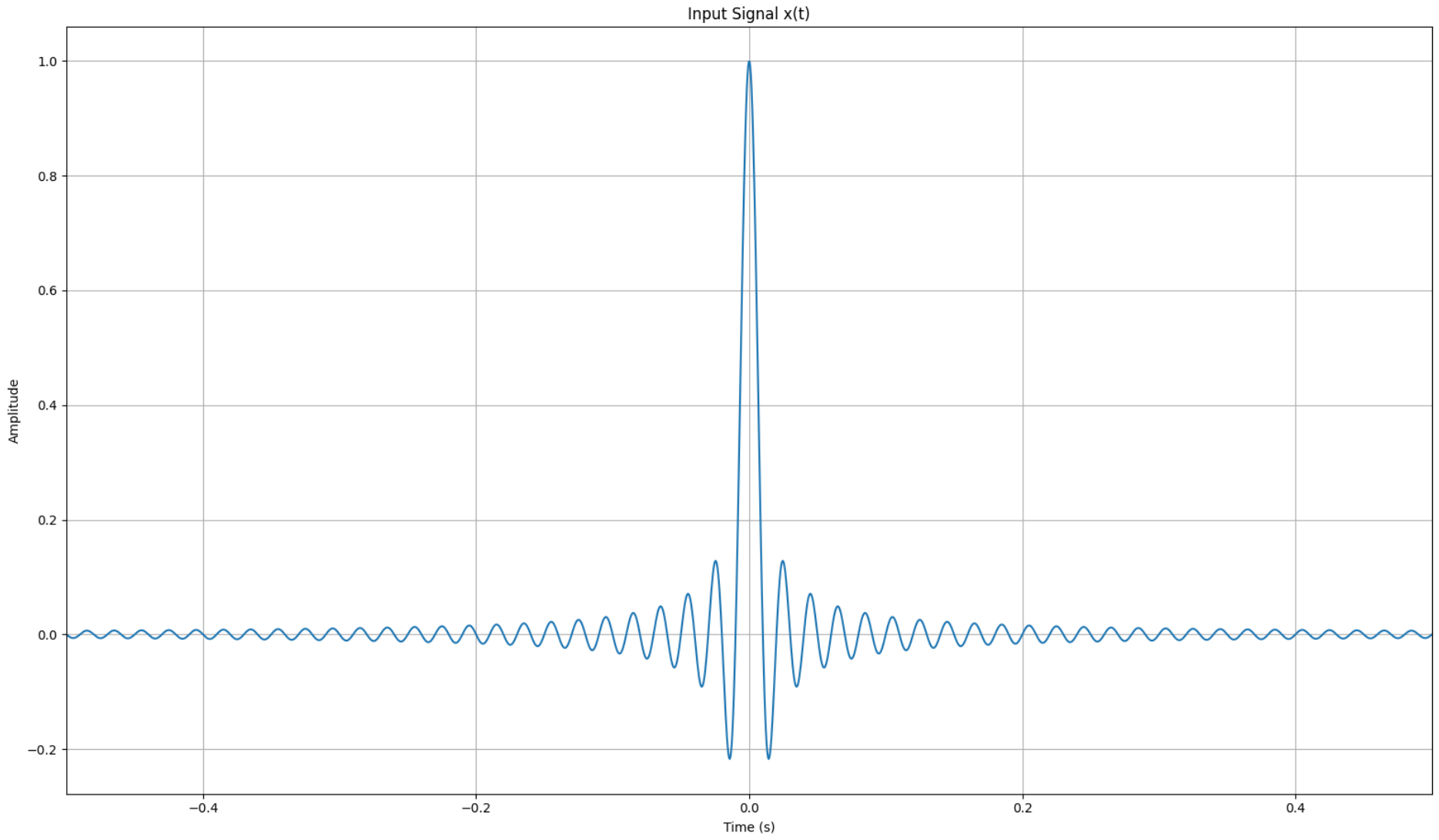
$$B = 2F \tag{9}$$

# Nyquist Rate

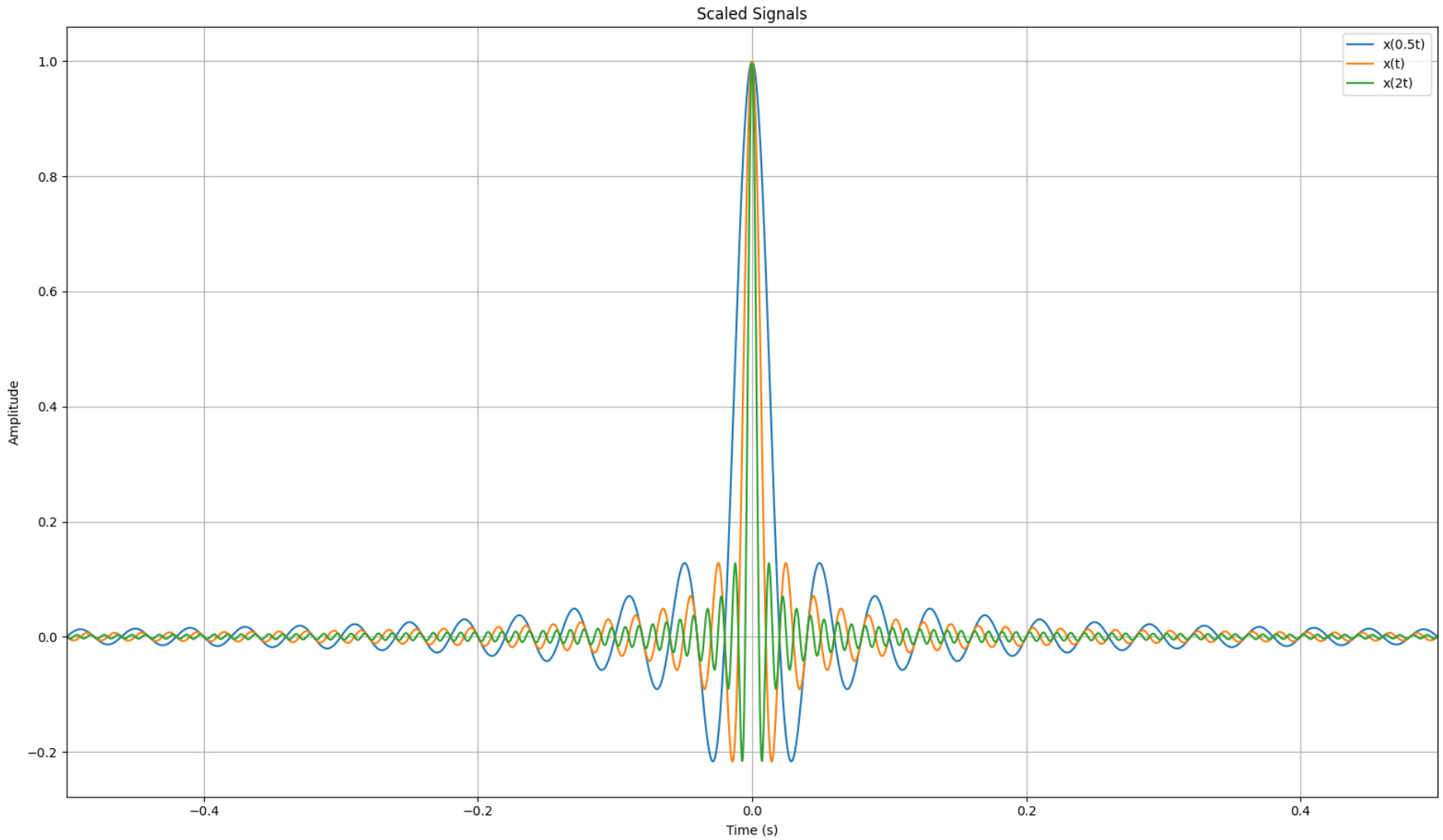
Finding Nyquist rate from definition.

$$2 \times B = 2 \times 2F = 4F \tag{10}$$

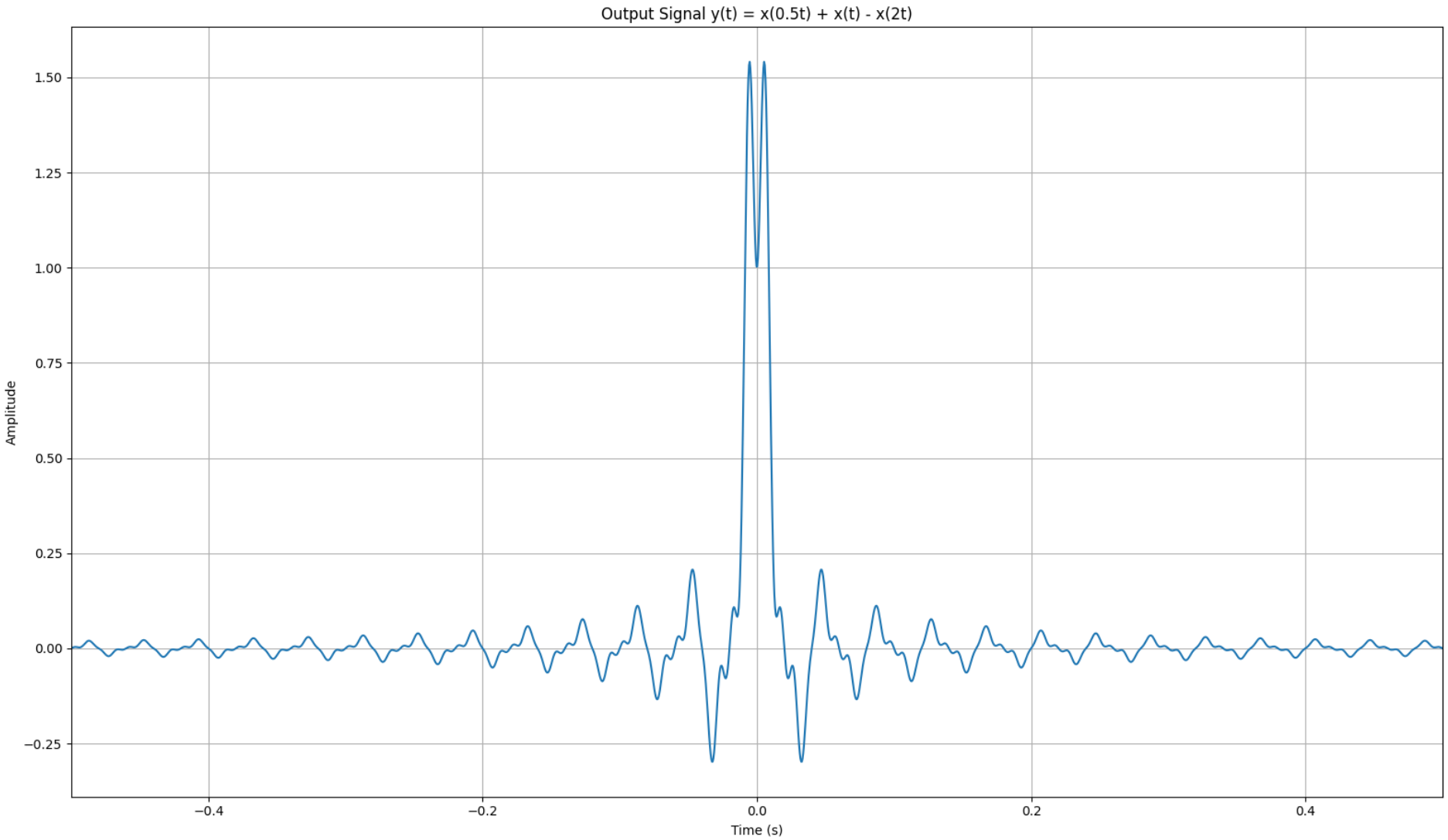
# Plot-x(t)


Input Signal x(t)
Amplitude
Time (s)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.2
0.0
0.2
0.4

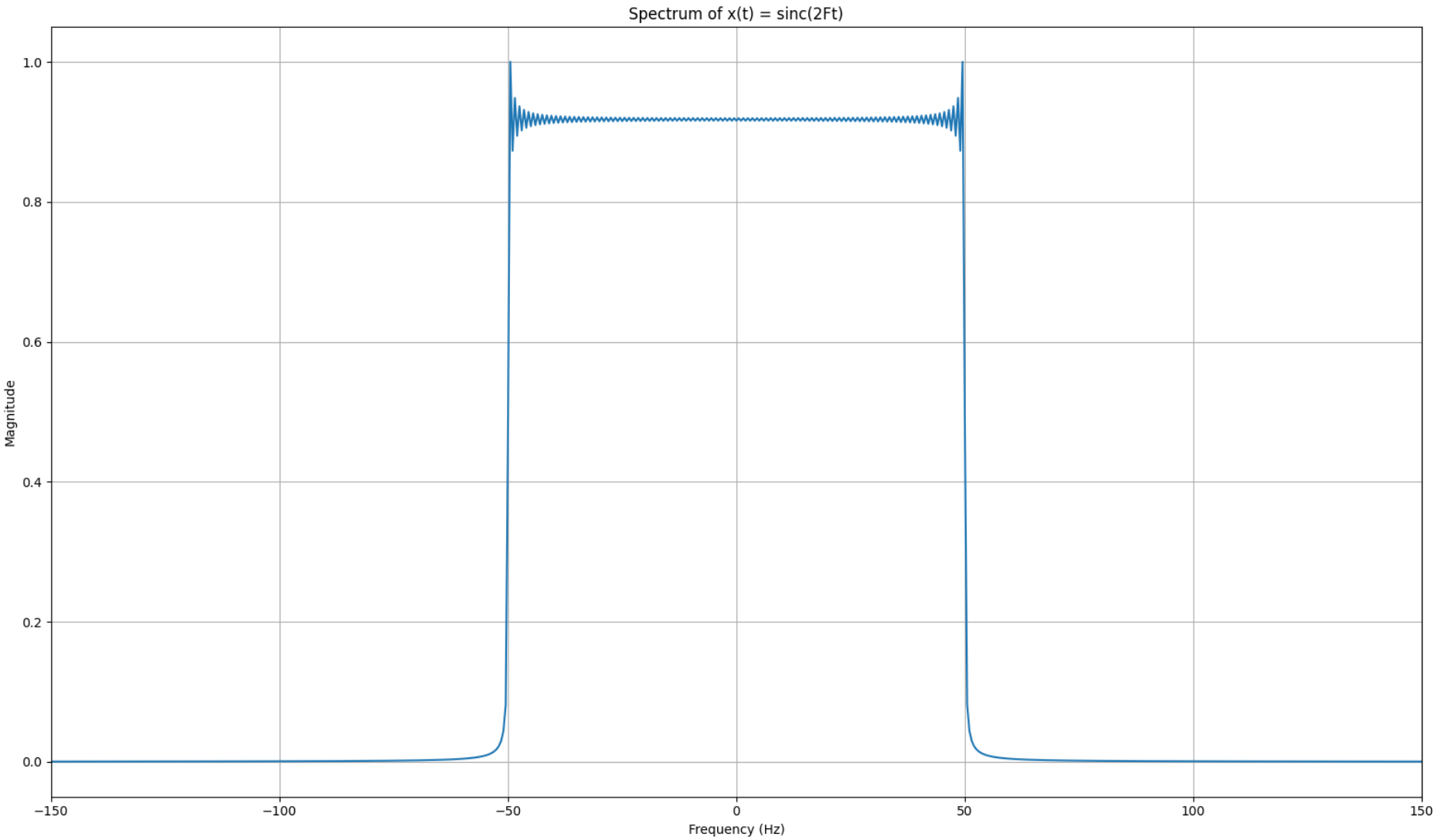
# Plot-y(t)


Scaled Signals
x(0.5t)
x(t)
x(2t)
Amplitude
Time (s)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.2
0.0
0.2
0.4

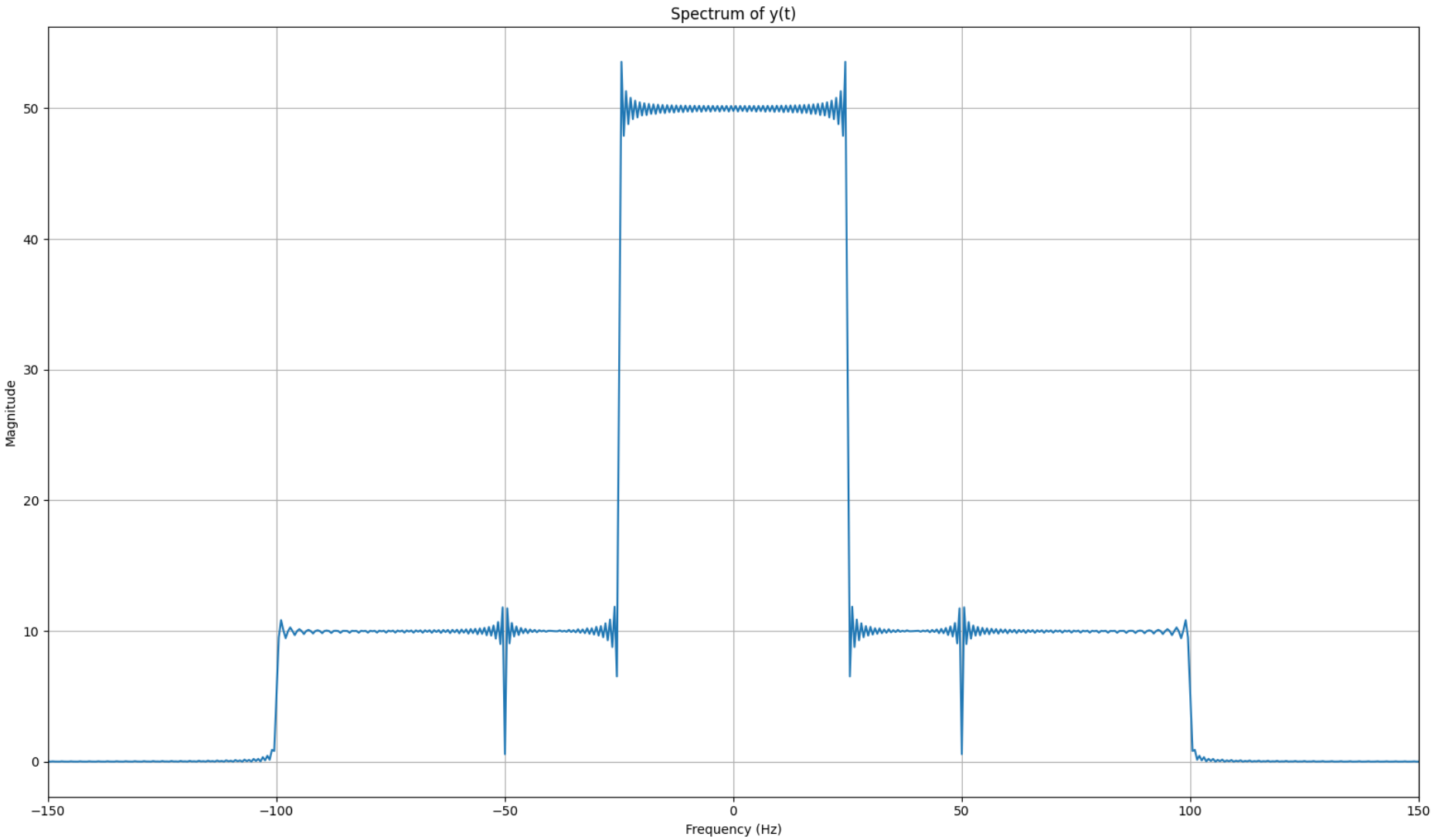
# Plot-y(t)


Output Signal y(t) = x(0.5t) + x(t) - x(2t)
Amplitude
Time (s)
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.4
−0.2
0.0
0.2
0.4

# Spectrum of x(t)



# Spectrum of y(t)



# Python Code 1

```
import numpy as np
import matplotlib.pyplot as plt
Fs = 2000
T = 2
t = np.linspace(-T/2, T/2, Fs*T)
F = 50
x = np.sinc(2 * F * t)

def scale_signal(x, t, scale):
    t_scaled = t * scale
    return np.interp(t_scaled, t, x, left=0, right=0)

x_05 = scale_signal(x, t, 0.5)
x_1 = x
x_2 = scale_signal(x, t, 2)
y = x_05 + x_1 - x_2
```

# Python Code 1

```
def compute_fft(signal, Fs):
    N = len(signal)
    X = np.fft.fftshift(np.fft.fft(signal))
    f = np.fft.fftshift(np.fft.fftfreq(N, d=1/Fs))
    return f, np.abs(X)

f_x, X = compute_fft(x, Fs)
X = X / np.max(X)

plt.figure()
plt.plot(f_x, X)
plt.title(Spectrum of x(t) = sinc(2Ft))
plt.xlabel(Frequency (Hz))
plt.ylabel(Magnitude)

plt.xlim(-150, 150)
plt.grid()
plt.show()
```

# Python Code

```
f, Y = compute_fft(y, Fs)

def estimate_bandwidth(f, spectrum, threshold=0.05):
    spectrum = spectrum / np.max(spectrum)
    indices = np.where(spectrum > threshold)[0]
    if len(indices) == 0:
        return 0
    max_freq = np.max(np.abs(f[indices]))
    return max_freq
bandwidth = estimate_bandwidth(f, Y)
nyquist_rate = 2 * bandwidth
print(Estimated Bandwidth:, bandwidth, Hz)
print(Estimated Nyquist Rate:, nyquist_rate, Hz)
plt.figure()
plt.plot(f, Y)
plt.title(Spectrum of y(t))
plt.xlabel(Frequency (Hz))
plt.ylabel(Magnitude)
plt.xlim(-150, 150)
```

# Python Code 2

```
import numpy as np
import matplotlib.pyplot as plt

Fs = 2000
T = 2
t = np.linspace(-T/2, T/2, Fs*T)
F = 50
x = np.sinc(2 * F * t)

def scale_signal(x, t, scale):
    t_scaled = t * scale
    return np.interp(t_scaled, t, x, left=0, right=0)

x_05 = scale_signal(x, t, 0.5)
x_1 = x
x_2 = scale_signal(x, t, 2)
y = x_05 + x_1 - x_2
```

# Python Code 2

```
plt.figure()
plt.plot(t, x)
plt.title(Input Signal x(t))
plt.xlabel(Time (s))
plt.ylabel(Amplitude)
plt.xlim(-0.5, 0.5)
plt.grid()
plt.show()

plt.figure()
plt.plot(t, x_05, label=x(0.5t))
plt.plot(t, x_1, label=x(t))
plt.plot(t, x_2, label=x(2t))
plt.title(Scaled Signals)
```

# Python Code 2

```
plt.title(Scaled Signals)
plt.xlabel(Time (s))
plt.ylabel(Amplitude)
plt.legend()
plt.xlim(-0.5, 0.5)
plt.grid()
plt.show()

plt.figure()
plt.plot(t, y)
plt.title(Output Signal y(t) = x(0.5t) + x(t) - x(2t))
plt.xlabel(Time (s))
plt.ylabel(Amplitude)
plt.xlim(-0.5, 0.5)
plt.grid()
plt.show()
```